

GALESBURG, IL, USA

WMO: 722089

Lat: 40.933N

Lon: 90.433W

Elev: 764

StdP: 14.29

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 94959

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.1	(c) 1.3	(d) -9.3	(e) 3.4	(f) -2.3	(g) -5.4	(h) 4.2	(i) 2.7	(j) 28.2	(k) 26.7	(l) 25.7	(m) 23.1	(n) 5.8	(o) 290	(p) 0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	91.4	75.1	89.8	74.5	87.1	73.1	78.4	87.7	76.9	86.1	75.4	83.9	8.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	137.9	83.9	73.9	130.0	82.1	72.6	124.3	80.6	42.6	88.0	40.8	85.9	39.3	84.2	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.4	(b) 20.6	(c) 18.3	(e) -12.2	(f) 97.1	(g) 6.5	(h) 7.1	(i) -16.9	(j) 102.2	(k) -20.7	(l) 106.3	(m) -24.4	(n) 110.3	(o) -29.1	(p) 115.4		
(4) 24.4	(5) 20.6	(6) 18.3	(7) -12.2	(8) 97.1	(9) 6.5	(10) 7.1	(11) -16.9	(12) 102.2	(13) -20.7	(14) 106.3	(15) -24.4	(16) 110.3	(17) -29.1	(18) 115.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	23.9	26.5	40.3	51.7	62.7	72.1	74.7	72.1	66.1	53.3	40.5	29.2
	DBStd	20.30	12.47	12.49	11.72	9.53	8.57	5.61	5.80	5.79	7.73	9.97	10.28	11.05
	HDD50	2960	811	661	347	93	8	0	0	0	1	78	314	647
	HDD65	6008	1275	1078	769	409	148	9	3	9	79	383	736	1110
	CDD50	3405	0	4	47	142	401	662	766	686	485	181	29	2
	CDD65	980	0	0	4	9	76	221	305	230	113	22	0	0
	CDH74	9249	0	0	32	120	710	2023	3093	2041	1026	204	0	0
	CDH80	3195	0	0	4	15	183	692	1219	706	328	48	0	0
(14) Wind	WSAvg	7.7	9.2	9.1	9.3	9.8	8.1	6.8	5.2	4.9	5.8	7.5	8.6	8.2
(15) Precipitation	PrecAvg	36.1	1.7	1.7	2.2	3.5	4.9	4.1	4.0	3.9	3.0	2.6	2.6	1.8
	PrecMax	50.1	3.6	5.0	4.8	6.2	10.6	8.1	11.1	8.7	6.6	5.9	5.9	3.8
	PrecMin	25.2	0.3	0.5	0.2	1.5	0.8	0.8	1.0	0.3	0.6	1.0	0.2	0.4
	PrecStd	5.8	0.9	1.1	1.4	1.3	2.6	2.2	2.4	2.0	1.8	1.4	1.5	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.7	64.1	78.9	82.0	89.6	92.7	97.3	94.8	91.0	85.8	71.8	60.1
		MCWB	50.2	55.3	63.1	62.9	70.4	74.9	77.0	77.2	71.5	69.6	57.7	56.9
	2%	DB	50.0	55.7	70.7	77.7	84.4	90.1	92.4	90.5	87.9	80.0	64.5	53.9
		MCWB	45.5	49.0	59.0	61.5	67.8	74.7	76.6	75.0	72.8	64.8	56.3	49.6
	5%	DB	44.8	48.5	64.0	73.2	81.4	87.5	89.8	87.5	84.0	74.4	61.1	48.4
		MCWB	41.2	44.2	55.6	59.5	65.4	72.5	75.5	73.8	70.7	63.4	54.0	45.2
	10%	DB	39.3	43.1	57.6	69.6	78.1	84.1	86.5	83.8	80.8	69.8	56.1	44.6
		MCWB	36.4	39.0	50.3	57.0	64.0	71.3	74.1	72.6	67.6	60.6	50.0	41.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	57.6	64.8	66.9	74.2	78.9	81.1	80.5	76.8	72.3	61.7	58.6
		MCDB	55.7	62.2	75.9	76.5	84.7	88.5	89.7	90.2	86.6	81.4	66.9	60.8
	2%	WB	46.2	50.3	60.4	64.1	71.2	76.8	79.1	77.7	74.6	68.9	58.2	50.2
		MCDB	48.9	54.4	68.4	73.1	81.6	85.9	88.4	86.9	83.4	76.2	63.2	52.4
	5%	WB	41.1	44.4	57.0	61.6	68.8	75.1	77.6	75.8	72.7	65.8	54.6	45.7
		MCDB	44.9	48.3	63.5	70.8	77.0	84.2	87.0	83.7	80.3	70.9	59.9	48.0
	10%	WB	36.9	39.4	51.2	58.7	66.6	73.3	75.9	73.9	70.7	62.2	50.5	41.1
		MCDB	39.4	42.8	57.8	67.5	74.1	81.5	84.6	80.7	77.6	68.2	56.0	44.3
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	17.1	18.7	21.4	20.9	20.1	20.2	20.5	22.4	20.5	18.1	14.6
		MCDBR	19.6	24.4	24.6	26.7	24.4	22.8	21.8	23.5	23.7	25.2	23.3	19.5
		MCWBR	15.9	17.7	15.3	14.8	11.6	10.7	9.5	10.7	10.7	12.8	15.7	16.0
	5% WB	MCDBR	19.1	22.5	22.3	23.5	19.8	19.4	19.5	19.8	20.0	20.1	20.8	19.1
		MCWBR	16.2	17.3	15.3	14.7	11.1	10.3	10.4	10.7	10.0	13.1	15.6	17.0
(40) Clear-Sky Solar Irradiance	taub		0.289	0.307	0.337	0.387	0.419	0.426	0.443	0.427	0.394	0.348	0.303	0.290
	taud		2.441	2.391	2.401	2.264	2.237	2.269	2.201	2.248	2.310	2.440	2.507	2.453
	Ebn at Noon		277	288	290	281	273	270	264	265	266	269	270	267
	Edh at Noon		25	30	34	42	43	42	45	42	36	29	23	23
(44) All-Sky Solar Radiation	RadAvg		567	876	1184	1496	1746	1963	1963	1727	1473	982	653	478
	RadStd		66	84	85	127	127	167	117	109	113	111	54	59

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page